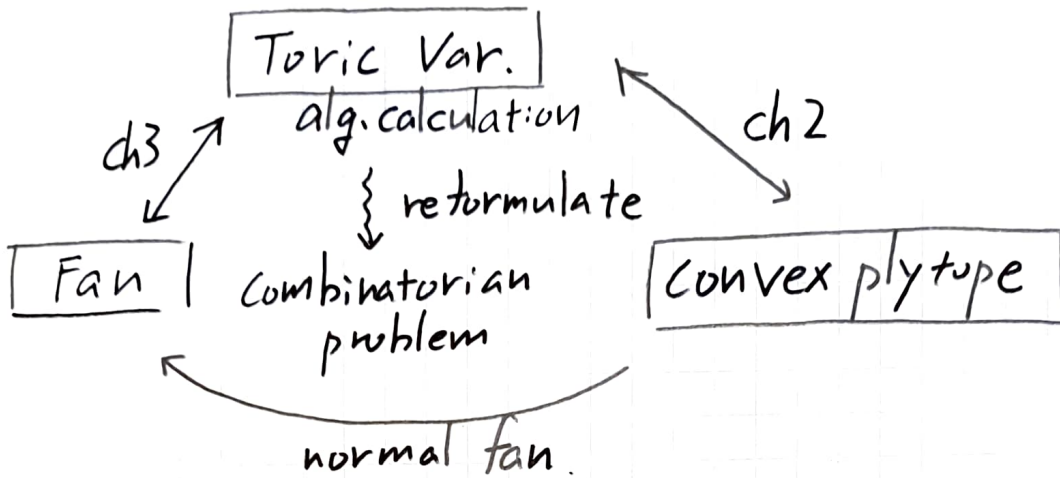


§ 3.4 Complete & proper (+ (Weil) divisor)

Motivation & Review



Today we focus on $\boxed{\text{Toric var}} \longleftrightarrow \boxed{\text{Fan}}$

$\boxed{\text{Fan } \Sigma (\subseteq N_{\mathbb{R}})}$

Q Collection of s.c.r.p.c w/
 $\begin{cases} \cdot \text{face of } \sigma \in \Sigma \\ \cdot \sigma_1 \cap \sigma_2 \in \Sigma \end{cases}$

$\boxed{\text{Toric Var. } X_{\Sigma}}$

Q irred. var containing a torus
 $T_N \simeq (\mathbb{C}^*)^n$ as a Zariski open
 s.t T_N action on itself
 extends to an alg. action
 of T_N on X_{Σ} .

Fan

Thm 3.1.5

normal, separated

Sumihiko

(Thm 3.1.7, Cor 3.1.8)

toric var

Hausdorff

(ideg)

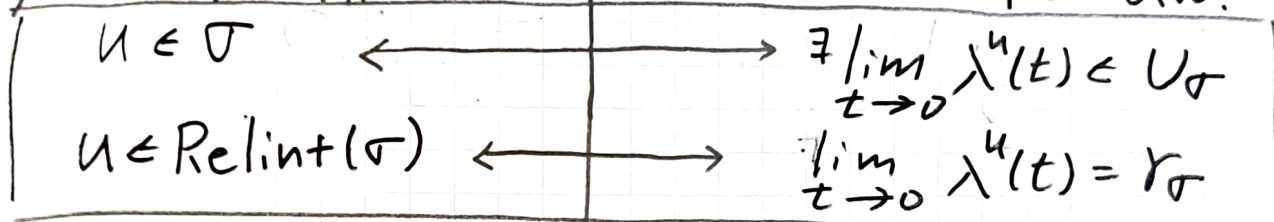
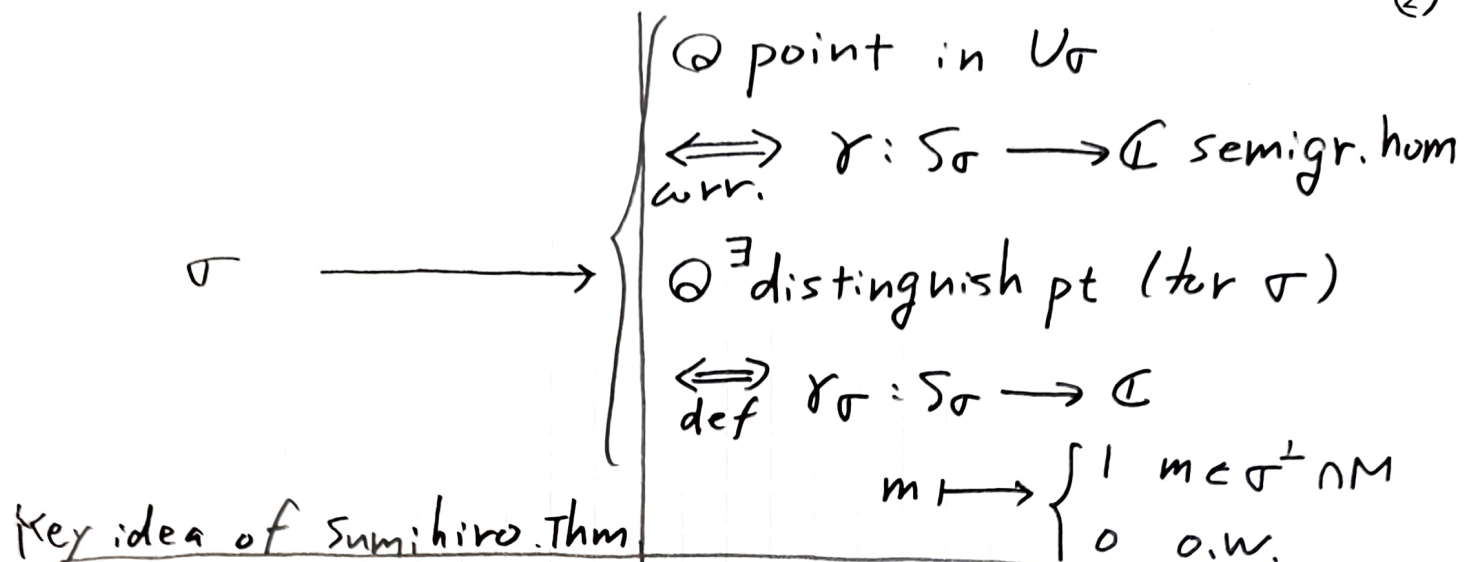
σ

take a some kind of
dual twice.

$U_{\sigma} = \text{Spec}(\mathbb{C}[\sigma^{\vee} \cap M])$

!!
So : semigr

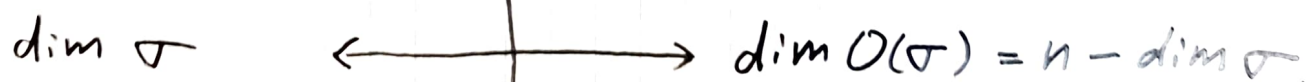
glue X_{Σ}



Cone-Orbit Corresponding



Rmk $V(\sigma) := \overline{O(\sigma)}$: toric var
 $w/ O(\sigma)$: torus
 $\text{Hom}(\sigma^\perp \cap M, \mathbb{C}^*)$



What is Weil divisor on toric var?

We need what's $\left\{ \begin{array}{l} \cdot \text{ prime div.} \\ \cdot \text{ valuation} \\ \cdot \text{ principal.} \end{array} \right.$

$P \in \Sigma(1)$
1-dim cone

$$\xrightarrow{\quad} \boxed{O(P) : T_N \text{ inv. prime div}}^{(3)}$$

$\parallel$
 D_P

$$w/ \boxed{\begin{array}{l} \nu_P : \mathbb{C}(X_\Sigma)^* \longrightarrow \mathbb{Z} \\ \chi^m \longmapsto \langle m, \nu_P \rangle \end{array}}$$

$$\nearrow \nu_P : \text{min. gen} \in P \cap N$$

We can regard characters as rational functions

($\odot$) $T_N \subset X_\Sigma$ Zariski open

$$\Rightarrow \boxed{\text{div}(\chi^m) = \sum_{P \in \Sigma(1)} \nu_P(\chi^m) D_P}$$

How to calculate $Cl(X_\Sigma)$

Thm 4.13

$$\begin{array}{ccccccc} (0 \rightarrow) M & \longrightarrow & \text{Div}_{T_N}(X_\Sigma) & \longrightarrow & Cl(X_\Sigma) & \longrightarrow & 0 \\ * & & & & \text{class gr.} & & \\ m & \longmapsto & \text{div}(\chi^m) & & & & \\ & & D & \longmapsto & [D] \text{ divisor class} & & \end{array}$$

$*$: $\{\nu_P | P \in \Sigma(1)\}$ spans $N_{\mathbb{R}}$
(i.e. X_Σ has no torus factor)

How to use this

$$\begin{array}{l} M \longrightarrow \text{Div}_{T_N}(X_\Sigma) \\ \leadsto A : \mathbb{Z}^r \longrightarrow \mathbb{Z}^r \\ \parallel \\ (u_1 \dots u_r) \end{array}$$

$(u_1 \dots u_r)$

ray generator

, then $Cl(X_\Sigma) \simeq \text{Coker } A$

example $\rightarrow$ see § 4.1

Fan

Toric Var.

④

smooth $\xleftrightarrow{\text{Thm 3.1.19(a)}} \text{smooth}$
 (def $\iff \forall \sigma : \text{smooth}$
 i.e. gen by $\mathbb{Z}$ basis of $N/\mathbb{R}$)

simplicial $\xleftrightarrow{\text{Thm 3.1.19(b)}} \mathbb{Q}$ -factorial
 (def $\iff \forall \sigma : \text{simplicial}$
 i.e. gen by $\mathbb{R}$ basis of $N_{\mathbb{R}}/\mathbb{R}$)
 (toric $\iff$ Orbifold)
 i.e. has only finite quot. sing.


complete $\xleftrightarrow{\text{Thm 3.1.19(c)}} \text{compact (classical)}$
 (def $\iff \text{support } |\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma = N_{\mathbb{R}}$)

relative var.

$\mathbb{Z}$ -linear compatible with fans $\longleftrightarrow$ Toric morphism
 i.e. maps torus into torus

e.g. star subdivision $\longleftrightarrow$ blow up.

Today

complete $\xleftrightarrow{\text{Thm 3.4.1}} \text{cpt (classical)}$

$\iff$ complete (alg. geom)

Criterion 1

$$\exists \lim_{t \rightarrow 0} \lambda^u(t) \cdot \text{in } X_{\Sigma} \quad \forall u \in N$$

GAGA

proper $\xleftrightarrow{\text{Thm 3.4.7}} \text{proper (classical)}$

$\iff$ proper (alg. geom)

SGA1

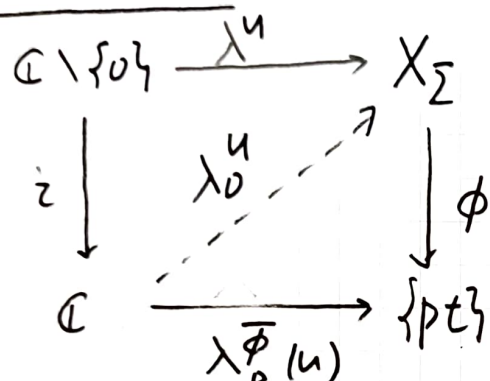
(Prop 3.2 of Exp XII) $\hookrightarrow$ univ. closed

criterion 2
 If $\exists \lim_{t \rightarrow 0} \lambda^{\bar{\varphi}(u)}(t) \in X_{\Sigma'}$,
 then $\exists \lim_{t \rightarrow 0} \lambda^u(t) \in X_{\Sigma}$

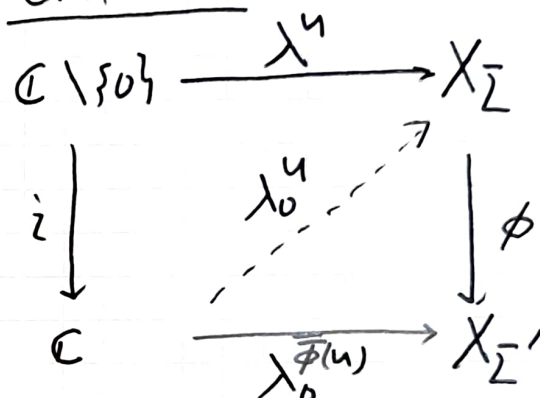
why relative?

$\phi: X \rightarrow Y$: proper, if $Y = \{pt\}$, then we define
 X is complete (alg. geom)

Criterion 1



Criterion 2



If we replace
 $\begin{cases} \mathbb{C} \setminus \{0\} \rightarrow \text{Spec } K & K: \text{field of frac.} \\ \mathbb{C} \rightarrow \text{Spec } R & R: \text{DVR} \end{cases}$
 $\Rightarrow$ valuative criterion for
 properness
 (Hart. II Thm 4.7, Ex 4.11)

@ Proof for compactness

Thm 3.4.1

- (a) X_{Σ} : cpt (classical)
- (b) $\exists \lim_{t \rightarrow 0} \lambda^u(t) \in X_{\Sigma} \quad (\forall u \in N)$
- (c) Σ : complete.

(6)

Rmk X_Σ is Hausdorff & each U_σ : metric
 $\Rightarrow X_\Sigma$: sequentially cpt.

<proof>

(a) $\Rightarrow$ (b)

Fix $u \in N$. Given seq $\{t_k\} \subset \mathbb{C}^*$ s.t. $t_k \rightarrow 0$

Consider $\{\lambda^u(t_k)\} \subset X_\Sigma$

cptness $\rightarrow \{\lambda^u(t_k)\}$ has subseq s.t. converge

$$\text{Let } \gamma = \lim_{k \rightarrow \infty} \lambda^u(t_k) \in X_\Sigma$$

$\uparrow$ after taking subseq.

Now we can assume

$\gamma \in U_\sigma$ for some $\sigma \in \Sigma$ (because $X_\Sigma = \bigcup_{\sigma \in \Sigma} U_\sigma$)

Take $m \in \sigma^\vee \cap M$, χ^m is regular fun. on U_σ ,
 thus continuous

$$\Rightarrow \chi^m(\gamma) = \lim_{k \rightarrow \infty} \chi^m(\lambda^u(t_k)) = \lim_{k \rightarrow \infty} t_k^{\langle m, u \rangle}$$

since $t_k \rightarrow 0$, thus $\langle m, u \rangle \geq 0$ for all $m \in \sigma^\vee \cap M$

$$\Rightarrow \langle m, u \rangle \geq 0 \text{ for all } m \in M$$

$$\Rightarrow u \in (\sigma^\vee)^\vee = \sigma$$

$$\Rightarrow \lim_{t \rightarrow 0} \lambda^u(t) \in U_\sigma \subseteq X_\Sigma$$

by Prop 3.2.2

(a) $\Rightarrow$ (b) OK

(b) $\Rightarrow$ (c) for any $u \in N$.

$$\exists \lim_{t \rightarrow 0} \lambda^u(t) \in U_\sigma \Rightarrow u \in \sigma \cap N$$

for $\exists \sigma \in \Sigma \Rightarrow$ every lattice pt

is contained in a cone of Σ

$$\Rightarrow |\Sigma| = N_{\mathbb{R}}$$

(b) $\Rightarrow$ (c) OK

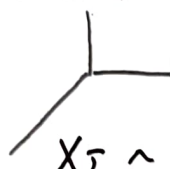
(c) $\Rightarrow$ (a) (Sketch)

(7)

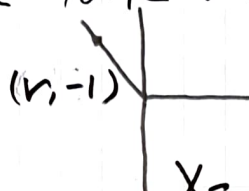
By induction on $n = \dim N_{\mathbb{R}}$

Q $n=1$ The only complete fan Σ is $\overleftarrow{\mathbb{R}_{\leq 0} \{0\} \mathbb{R}_{\geq 0}}$
 $X_{\Sigma} = \mathbb{P}^1 \simeq S^1$ cpt OK

c.f. $n=2$ sm cpt toric var.



$$X_{\Sigma} \simeq \mathbb{P}^2$$



$$X_{\Sigma} \simeq \mathcal{H}_r : \text{Hirzebruch} \\ = \mathbb{P}(\mathcal{O} \oplus \mathcal{O}(r))$$

$$r=0 \\ \mathbb{P}^1 \times \mathbb{P}^1$$

Rmk

- Toric var $\Rightarrow$ rational
- min. rat. surf $\Rightarrow \mathbb{P}^2$ or $\mathcal{H}_r$

Q assume $\dim < n$ OK

let $\{\gamma_k\} \subset X_{\Sigma} : \text{seg}$

Since $X_{\Sigma} = \bigcup_{\text{fin}} \mathcal{O}(\mathcal{I})$, we can assume $\{\gamma_k\} \subset \mathcal{O}(\mathcal{I})$

for some $\mathcal{I} \in \Sigma$

we only need to get subseg at the end.

Case 1: $\mathcal{I} \neq \{0\}$

$$\overline{\mathcal{O}(\mathcal{I})} = X_{\text{star}(\mathcal{I})} : \dim \leq n-1$$

then using assumption of induction.

$\rightarrow$ OK.

Case 2: $\mathcal{I} = \{0\}$

$$\mathcal{O}(\mathcal{I}) = T_N \simeq \text{Hom}_{\mathbb{Z}}(M, \mathbb{C}^*)$$

Now we regard $\gamma \in T_N$ as $\gamma : M \rightarrow \mathbb{C}^*$

Key (logarithm map)

(8)

given $r: M \longrightarrow \mathbb{C}^*$. we define

$$L(r): M \longrightarrow \mathbb{R} \\ m \longmapsto \log |r(m)|$$

Note that $L(r) \in \text{Hom}_{\mathbb{Z}}(M, \mathbb{R}) \cong N_{\mathbb{R}}$,
so L defines $L: T_N \longrightarrow N_{\mathbb{R}}$

Now we suppose for $r \in T_N$

$L(r) \in -\sigma$ for some $\sigma \in \Sigma$

If $m \in \sigma^\vee \cap M$, then

$$\log |r(m)| = \langle m, L(r) \rangle \leq 0$$

$$\Rightarrow |r(m)| \leq 1$$

So we get $L(r) \in -\sigma \Rightarrow |r(m)| \leq 1$

Now $L(r_k) \in N_{\mathbb{R}} \ \& \ \bigcup_{\sigma \in \Sigma} -\sigma$ is also $N_{\mathbb{R}}$

taking subseq

$$\longrightarrow L(r_k) \in -\sigma \quad (\forall k)$$

$$\Rightarrow |r_k(m)| \leq 1 \quad (\forall m \in \sigma^\vee \cap M)$$

$$\Rightarrow r_k: M \longrightarrow \underbrace{B(0,1)}_{\text{cpt}}$$

$$\Rightarrow \exists \{r_{k_\ell}\}: \text{subseq}$$

Ex 3.4.3

converges to $r \in \bigcup \sigma$

using $\sigma^\vee \cap M$: finiteness
(Gordan)

(c) $\Rightarrow$ (a) ok

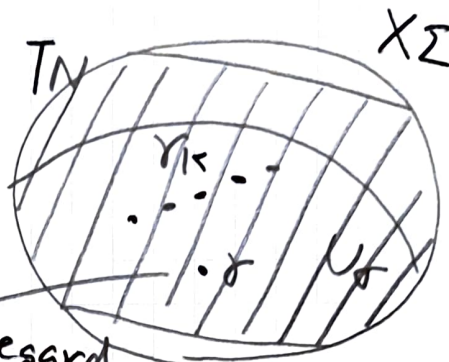
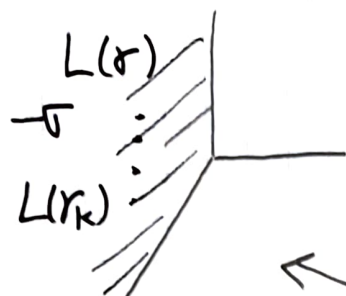
✕

Image

$$N = |\Sigma|$$

$$M = \text{Hom}_{\mathbb{Z}}(N, \mathbb{R})$$

(9)



$$L(r)(m) = \log |r(m)|$$

